

SentinelShield

Advanced Linux Security Hardening & Threat Audit Framework

Course Title: Operating Systems Lab

Course Code: CSE-402

Section: 231-D2

Course Teacher's Name: Md. Shoaib Alam

The Threat Audit Team: Expertise Driving Linux Security

Our team focused on Linux security, automation, and operating system concepts to build a practical academic security framework.



Chinmoy Debnath

ID-231902029

Dept of CSE

Green University of Bangladesh



Promod Chandra Das

ID-231902029

Dept of CSE

Green University of Bangladesh

Why SentinelShield?

Linux systems are often misconfigured. SentinelShield provides an Menu-driven automated, policy-based, and safer way to audit, validate, harden, and report Linux security settings..

Manual security checking is:

1. Time-consuming
2. Error-prone
3. Inconsistent



Main Goal?

To design a **Bash-based Linux security framework** that can:

1. Audit system security
2. Validate against SSH security policies
3. Safely apply SSH hardening
4. Roll back on failure
5. Generate security reports with score and logs



Security Workflow

Detect → Validate → Harden → Verify → Report

1

Detect

Find risky configurations.

2

Validate

Compare with policy (INI).

3

Harden

Apply secure settings (safe).

4

Verify

Validate config, then restart.

5

Report

Score + log + evidence



Technologies Used



Bash Script

Whiptail (menu-driven TUI)

INI configuration file

What is audited?

- ☐ World-writable files & directories
- ☐ Open / listening ports
- ☐ Running & enabled services
- ☐ Sticky-bit checks
- ☐ SUID / SGID binaries



✓ World-writable files & directories



✓ Open ports



✓ Running & enabled services



✓ Sticky-bit validation (1777)



✓ Sticky 1777



✓ SUID SGID binaries

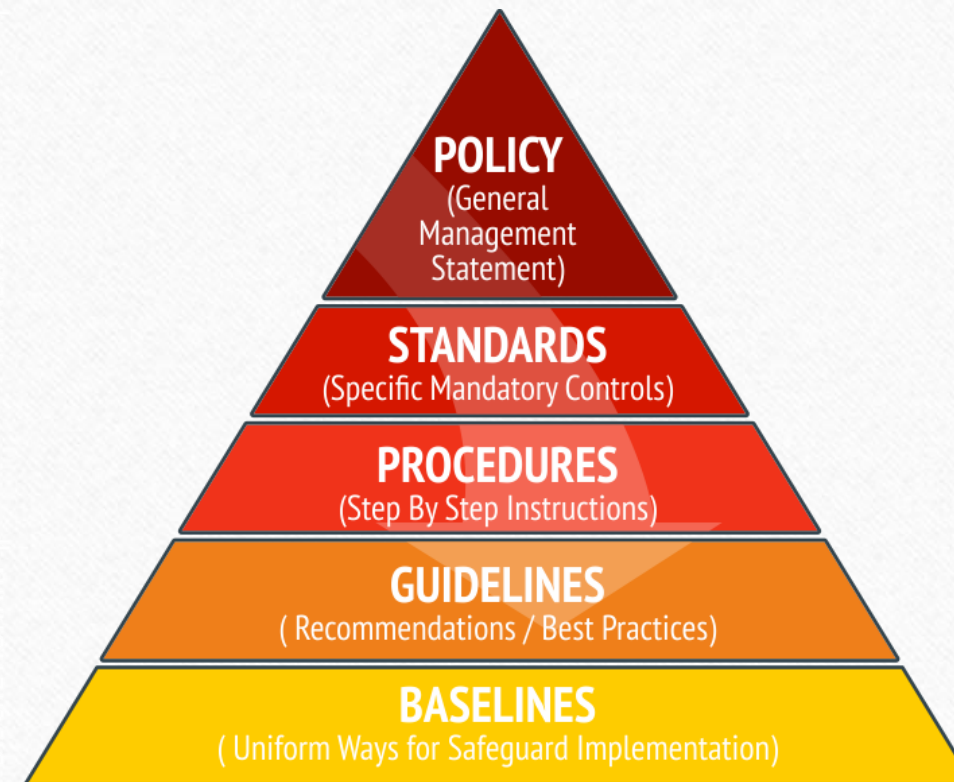
Policy-based Security?

sentinel_policy.ini

Example rules:
PermitRootLogin = no
PasswordAuthentication = no

Rule Logic

- Load sentinel_policy.INI
- Read current SSH config
- Compare current vs policy
- Show PASS or FAIL



SSH Hardening (Safe Design)

Hardening Steps.

1. Backup `/etc/ssh/sshd_config` before making changes
2. Load SSH policy values from `sentinel_policy.ini`
3. Apply `PermitRootLogin` and `PasswordAuthentication` settings
4. Validate the updated SSH configuration using `sshd -t`
5. Restart SSH if valid; otherwise restore the backup automatically



Report & Scoring System

Generated Report Includes

- System information
- Audit results
- SSH security status
- Open ports & services
- Security score (0–100)

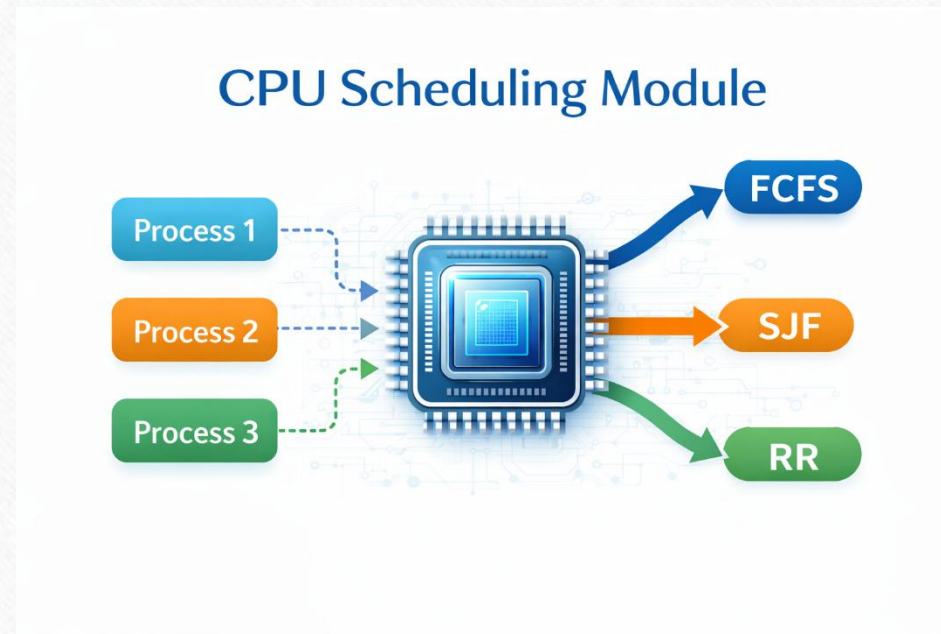
Scoring Logic

- World-writable issues → score deduction
- Policy mismatch → score deduction
- SSH service status → reported separately

CPU Scheduling Module

Algorithms Implemented

- FCFS
- SJF (Non-preemptive)
- Round Robin (Quantum)
- renice-based process priority adjustment



Output

- Waiting Time (WT)
- Turnaround Time (TAT)
- Gantt Chart
- Average WT & TAT

Conclusion & Future Scope?

Achievements

- Safe SSH hardening with rollback
- Policy-based decision making
- User-friendly menu interface
- Menu-driven automated Linux security audit



Future Enhancements

- Firewall (UFW) hardening
- CIS benchmark checks
- HTML / PDF reports
- Cron-based scheduled scans



**HAVE YOU ANY
QUESTION?**

